Keanu Natchev

514-929-6306, keanu.natchev@gmail.com, github.com/keanutan, linkedin.com/in/keanu-natchev/
English – French – Bulgarian

Education

Bachelor of Software Engineering **December 2022**
McGill University, Montreal, QC

Technical Skills

Languages: Java, TypeScript, C, Kotlin, Scala, Python, CUDA, JavaScript, OCaml, VHDL, ARM, BASH
Tools/OS: Git, Cucumber, Gradle, Maven, Heroku, Travis CI / MacOS, Windows, Linux, Ubuntu
Frameworks: React, Spring Boot, Vue.js
Cloud Services: AWS (ECS, Fargate, Lambda, DynamoDB, Aurora, RDS, S3, Cloudwatch, IAM, Cloudfront)
IDEs: Visual Studio Code, Eclipse, Adobe Dreamweaver

Work Experience

Software Development Engineer | Amazon **January 2023 – Present**

- Designed and implemented multiple cross-team technical initiatives, demonstrating strong ownership and collaboration skills.
- Delivered solutions across backend, frontend, and data engineering domains.
- Led critical incident response operations as primary on-call engineer through rapid issue identification and remediation, while establishing robust post-incident procedures.

Software Development Engineer Intern | Amazon **Summer 2022**
Supervisor and Floor Clerk | Shoppers Drug Mart **June 2016 - December 2019**

Personal Engineering Projects

Portfolio Website (keanunatchev.com) **July 2021**
Binary Search Tree Visualizer (available on GitHub) **December 2020**

University Engineering Projects

Novel Representation of Sign Language to Aid in Deaf Literacy **Winter – Fall 2022**
Capstone Design Project (ECSE 458)

- Collaborated with a **team of three** to design and develop a **React** web application aimed at improving literacy rates among deaf and hard-of-hearing individuals by creating a **novel sign language representation**.
- Contributed to all phases of the **project lifecycle**, including requirements elicitation, iterative design, web development, and documentation, fostering effective teamwork through asynchronous task management and group decision-making.
- Created and iterated on paper and digital prototypes, **incorporating feedback from advisors and deaf community consultants** to preserve the dynamism of sign language, including body language and facial expressions.

Operating System in C **January 2021 - March 2021**
Operating Systems (COMP 310 & ECSE 427)

- Developed a **C-based operating system** with **shell** and **kernel**, implementing **boot sequence** and **memory management** for RAM pages, frames, and page fault handling.

Data Structure and Algorithm Visualization Website **September 2020 - December 2020**
Software Engineering Practice (ECSE 428)

- Built a **React**-based website in an **8-person agile SCRUM team** to visualize sorting algorithms on data structures, using **Anime.js** for animations and **GitHub** for version control.
- Designed **Array** and **Doubly Linked List** visualization pages.

Lego EV3 Mindstorms Robot **September 2020 - December 2020**
Design Principles and Methods (ECSE 211)

- Developed a **Java**-based robot in a **6-person team** to navigate a virtual obstacle course in **Webots**, designing hardware in **LeoCAD** and implementing an optimized **Localization class with multithreading**.

Event Registration System **January 2020 – April 2020**
Introduction to Software Engineering (ECSE 321)

- Created a **Vue.js**-based event registration website with **Java Spring Boot REST API backend**, using **UML Lab** for modeling, **Heroku** for database deployment, and **Travis CI** for continuous integration.